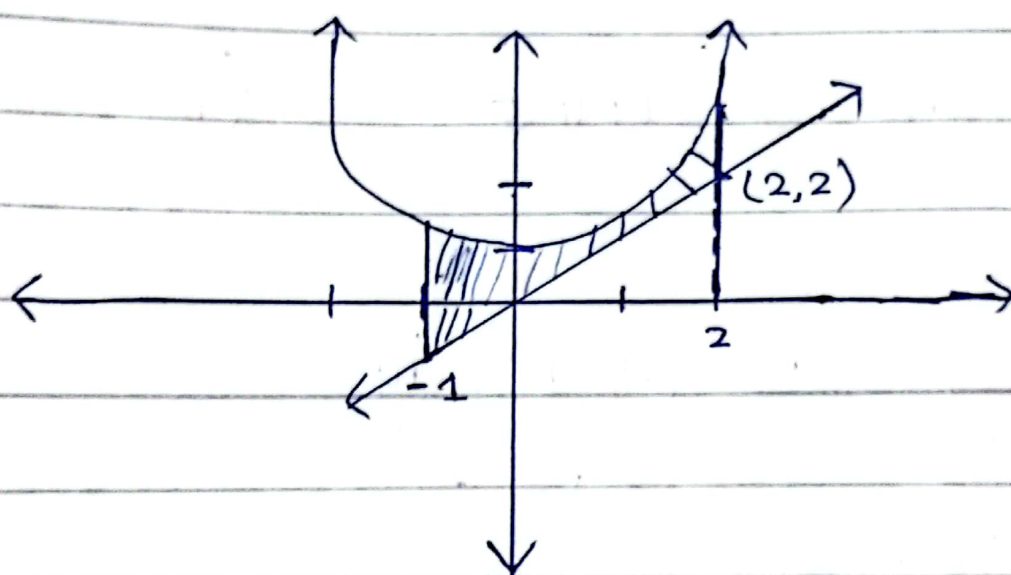


Exercise 6.1
1-4. Find area of Shaded region.

1. Along x-axis



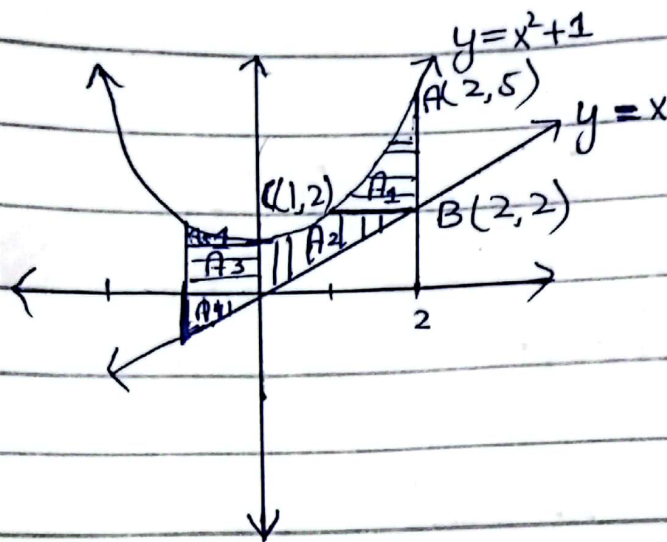
$$A = \int_{-1}^2 [(x^2 + 4) - (x)] dx = \int_{-1}^2 (x^2 - x + 4) dx$$

$$= \left[\frac{x^3}{3} - \frac{x^2}{2} + 4x \right]_{-1}^2$$

$$= \left[\frac{8}{3} - \frac{4}{2} + 8 \right] - \left[-\frac{1}{3} - \frac{1}{2} - 4 \right]$$

$$= \frac{16}{1} - \left(-\frac{11}{6} \right) = \frac{9}{2}$$

Along y-axis:



$$x = \pm\sqrt{y-1}, \quad x = y, \quad x = 2, \quad x = -1$$

$$A_T = A_1 + A_2 + A_3 + A_4 + A_5 \quad \text{--- (i)}$$

A_1 :

$$\begin{aligned} A_1 &= \int_{y=2}^{y=5} [2 - (\sqrt{y-1})] dy \\ &= \left(2y - \frac{2(y-1)^{3/2}}{3} \right) \Bigg|_2^5 \\ &= \left[2(5) - \frac{2(4)^{3/2}}{3} \right] - \left[2(2) - \frac{2(1)^{3/2}}{3} \right] \\ &= \left[10 - \frac{16}{3} \right] - \left[4 - \frac{2}{3} \right] \\ &= \frac{14}{3} - \frac{10}{3} = \frac{4}{3} \end{aligned}$$

H₂

$$\begin{aligned} A_2 &= \int_{y=0}^{y=2} [y - \sqrt{y-1}] dy \\ &= \left[\frac{y^2}{2} - \frac{2(y-1)^{3/2}}{3} \right]_0^2 \\ &= \frac{4}{2} - \frac{2}{3} - 0 \end{aligned}$$

$$A_2 = \frac{8}{6} = \frac{4}{3}$$

A₅:

$$\begin{aligned} A_5 &= \int_{y=1}^{y=2} [-\sqrt{y-1} - (-1)] dy \\ &= \int_1^2 (1 - \sqrt{y-1}) dy \\ &= \left(y - \frac{2(y-1)^{3/2}}{3} \right) \Big|_1^2 \\ &= \left[2 - \frac{2}{3} \right] - [1 - 0] \end{aligned}$$

$$A_5 = \frac{4}{3} - 1 = \frac{1}{3}$$

A₄:

$$\begin{aligned} A_4 &= \int_{y=-1}^{y=0} [y - (-1)] dy = \int_{-1}^0 (y+1) dy \\ &= \left[\frac{y^2}{2} + y \right]_{-1}^0 = 0 - \left[\frac{1}{2} - 1 \right] = \frac{1}{2} \end{aligned}$$

A₃:

$$\begin{aligned} A_3 &= \int_{y=0}^{y=1} [0 - (-1)] dy \\ &= \int_0^1 1 dy = y \Big|_0^1 = 1 \end{aligned}$$

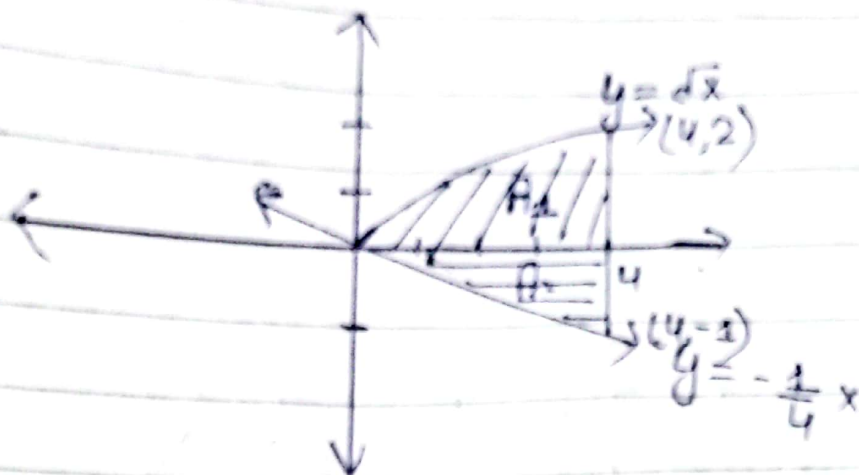
Then - (i) becomes

$$\begin{aligned} A_T &= \frac{4}{3} + \frac{4}{3} + 1 + \frac{1}{2} + \frac{1}{3} \\ &= \frac{8+8+6+3+2}{6} = \frac{27}{6} = \frac{9}{2} \end{aligned}$$

2. Along x-axis

$$\begin{aligned} A &= \int_{x=0}^{x=4} \left[\sqrt{x} - \left(\frac{1}{4}x \right) \right] dx \\ &= \int_0^4 \left(\sqrt{x} + \frac{1}{4}x \right) dx \\ &= \left(\frac{2x^{3/2}}{3} + \frac{x^2}{8} \right) \Big|_0^4 \\ &= \left[\frac{2}{3}(4)^{3/2} + 2 \right] - 0 \\ &= \frac{16}{3} + 2 = \frac{22}{3} \end{aligned}$$

Along y-axis



$$x = y^2, \quad x = -4y, \quad x = 4$$

$$A_T = A_1 + A_2 \quad \text{--- (i)}$$

A_1 :

$$\begin{aligned} A_1 &= \int_{y=0}^{y=2} [4 - y^2] dy \\ &= \left(4y - \frac{y^3}{3} \right) \Big|_0^2 \\ &= \left[8 - \frac{8}{3} \right] - 0 \\ &= \frac{16}{3} \end{aligned}$$

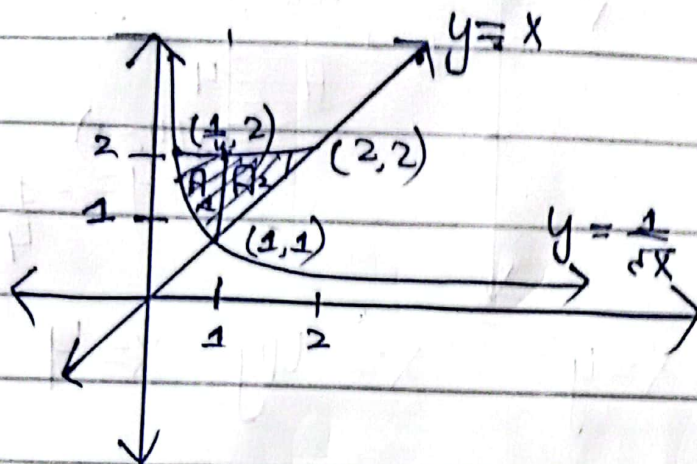
$$\begin{aligned} \textbf{A}_2: \quad A_2 &= \int_{-1}^0 (4 + 4y) dy \\ &= 4 \int_{-1}^0 (1 + y) dy \end{aligned}$$

$$= 4 \left[y + \frac{y^2}{2} \right]_{-1}^0 = 4 \left[0 - \left(-1 + \frac{1}{2} \right) \right]$$

Then, (i)

$$A_T = \frac{16}{3} + 2 = \frac{22}{3}$$

3)



$$y = x, \quad y = \frac{1}{\sqrt{x}}, \quad y = 2$$

$$A_T = A_1 + A_2 \quad \text{--- (i)}$$

$$2 = \frac{1}{\sqrt{x}}$$

A_1 :

$$A_1 = \int_{x=1/4}^{x=1} \left(2 - \frac{1}{\sqrt{x}} \right) dx$$
$$= \left[2x - 2\sqrt{x} \right]_{1/4}^1$$

$$= 2 \left[x - \sqrt{x} \right]_{1/4}^1$$

$$= 2 \left[0 - \frac{1}{4} + \frac{1}{2} \right]$$

$$A_1 = 2 \left[\frac{1}{4} \right] = \frac{1}{2}$$

A_2 :

$$A_2 = \int_{x=1}^{x=2} [2-x] dx$$

$$= \left[2x - \frac{x^2}{2} \right]_1^2$$

$$= \left[4 - 2 \right] - \left[2 - \frac{1}{2} \right]$$

$$= 2 - \frac{3}{2}$$

$$A_2 = \frac{1}{2}$$

Then (i) becomes

$$A_1 = \frac{1}{2} + \frac{1}{2} = 1$$

Along y-axis

$$A = \int_1^2 \left(y - \frac{1}{y^2} \right) dy$$

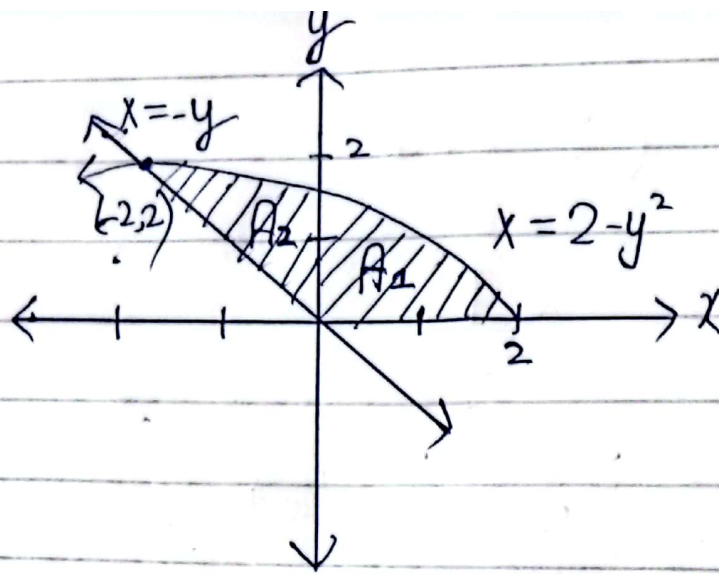
$$= \left[\frac{y^2}{2} - \frac{y^{-1}}{-1} \right]_1^2$$

$$= \left[\frac{y^2}{2} + \frac{1}{y} \right]_1^2$$

$$= 2 + \frac{1}{2} - \frac{1}{2} - 1$$

$$A = 1$$

4)



$$y = -x$$

$$y = \sqrt{2-x}$$

$$A_T = A_1 + A_2 \quad \text{--- (i)}$$

A_1 :

$$A_1 = \int_{x=0}^{x=2} \sqrt{2-x} \, dx$$

$$= (-1) \int_0^2 \sqrt{2-x} \, (-1) dx$$

$$= -\frac{2}{3} (2-x)^{3/2} \Big|_0^2$$

$$= -\frac{2}{3} [0 - 2^{3/2}]$$

$$A_1 = \frac{2^{5/2}}{3}$$

$$A_2 = \int_{x=-2}^{x=0} [\sqrt{2-x} - (-x)] \, dx$$

$$= \int_{-2}^0 (\sqrt{2-x} + x) \, dx$$

$$= \int_{-2}^0 \sqrt{2-x} \, dx + \int_{-2}^0 x \, dx$$

$$= -\frac{2}{3} (2-x)^{3/2} \Big|_{-2}^0 + \frac{x^2}{2} \Big|_{-2}^0$$

$$= -\frac{2}{3} [2^{3/2} - 4^{3/2}] + \frac{1}{2} [0 - 4]$$

$$= -\frac{2}{3} [2^{3/2} - 2^3] - 2$$

$$= -\frac{2}{3} \cdot 2^{3/2} [1 - 2^{3/2}] - 2$$

$$= -\frac{2^{5/2}}{3} [1 - 2^{3/2}] - 2$$

$$A_2 = -\frac{2^{5/2}}{3} + \frac{2^4}{3} - 2$$

Then (i) becomes

$$A_T = A_1 + A_2$$

$$= \frac{2^{5/2}}{3} - \frac{2^{5/2}}{3} + \frac{16}{3} - 2$$

$$\boxed{A_T = 10/3}$$

Along y-axis:

$$A = \int_{y=0}^{y=2} [(2-y^2) - (-y)] dy$$

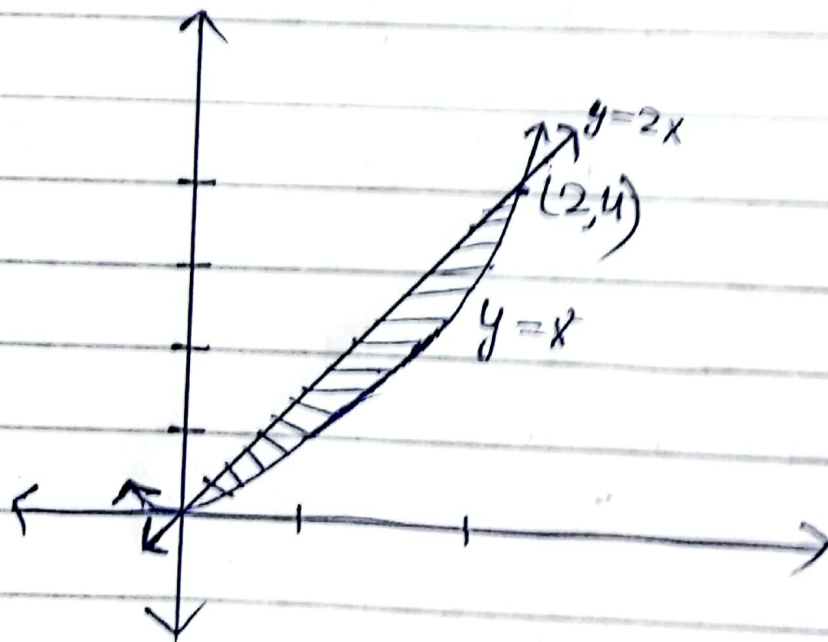
$$= \int_0^2 (-y^2 + y + 2) dy$$

$$= \left[-\frac{y^3}{3} + \frac{y^2}{2} + 2y \right]_0^2$$

$$= -\frac{8}{3} + 4 + 4$$

$$A = \frac{-16 + 12 + 24}{6} = \frac{20}{6} = \frac{10}{3}$$

5)



$$A = \int_{x=0}^{x=2} [2x - x^2] dx$$

$$= \left[2\frac{x^2}{2} - \frac{x^3}{3} \right]_0^2 = \left[2 - \frac{8}{3} \right] - 0$$

$$= 4 - \frac{8}{3} = \frac{4}{3}$$

Along y-axis

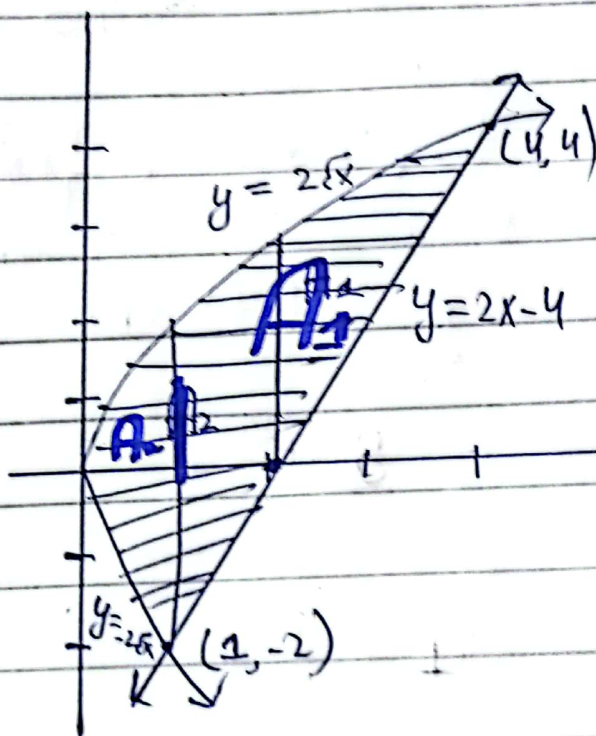
$$A = \int_{y=0}^{y=4} (\sqrt{y} - y/2) dy$$

$$= \left[\frac{2y^{3/2}}{3} - \frac{y^2}{4} \right]_0^4$$

$$= \frac{2}{3}(4)^{3/2} - \frac{(4)^2}{4} - 0$$

$$= \frac{16}{3} - 4 = \frac{4}{3}$$

Along x-axis



$$A_T = A_1 + A_2$$

$$A_1 = \int_1^4 [2\sqrt{x} - (2x - 4)] dx$$

$$= 2 \int_1^4 [\sqrt{x} - x + 2] dx$$

$$\begin{aligned}
&= 2 \left[\frac{x^{3/2}}{3/2} - \frac{x^2}{2} + 2x \right]_1^4 \\
&= 2 \left[\left(\frac{2(4)^{3/2}}{3} - \frac{16}{2} + 2(4) \right) - \left(\frac{2}{3} - \frac{1}{2} + 2 \right) \right] \\
&= 2 \left[\left(\frac{16}{3} - 8 + 8 \right) - \left(\frac{4-3+12}{6} \right) \right] \\
&= 2 \left[\left(\frac{32}{6} - 48 + 48 \right) - \frac{13}{6} \right] \\
&= 2 \left[\frac{32}{6} - \frac{13}{6} \right] \\
&= 2 \left(\frac{19}{6} \right) = \frac{19}{3}
\end{aligned}$$

$$\begin{aligned}
A_2 &= \int_{x=0}^{x=1} [2\sqrt{x} - (-2\sqrt{x})] dx \\
&= \int_0^1 4\sqrt{x} dx \\
&= \left[\frac{8}{3} x^{3/2} \right]_0^1 \\
&= \frac{8}{3} \left[\frac{1}{3} \right] = \frac{8}{3}
\end{aligned}$$

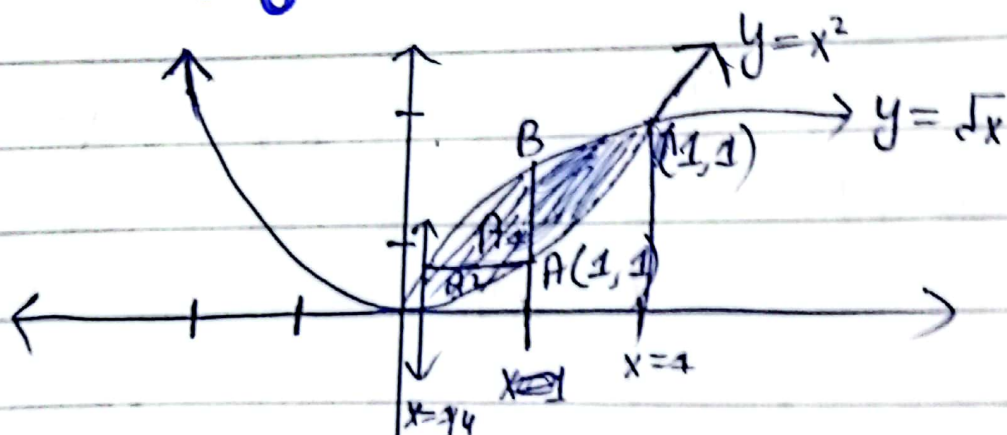
Then (i) becomes

$$A_T = \frac{19}{3} + \frac{8}{3} = \frac{27}{3} = 9$$

along y-axis

$$\begin{aligned}
 A &= \int_{y=-2}^{y=4} \left(y/2 + 2 - \frac{y^2}{4} \right) dy \\
 &= \left[\frac{y^2}{4} + 2y - \frac{y^3}{12} \right]_{-2}^4 \\
 &= \left[\frac{4^2}{4} + 8 - \frac{64}{12} \right] - \left[\frac{4}{4} - 4 + \frac{8}{12} \right] \\
 &= \left[12 - \frac{64}{12} \right] - \left[1 - 4 + \frac{8}{12} \right] \\
 &= \frac{144 - 64}{12} - \left[\frac{-36 + 8}{12} \right] \\
 &= \frac{80}{12} + \frac{28}{12} \\
 &= \frac{108}{12} = 9
 \end{aligned}$$

7) $y = x^2$, $y = \sqrt{x}$, $x = \frac{1}{4}$, $x = 1$



$$A = \int_{x=1/4}^{x=1} (\sqrt{x} - x^2) dx$$

$$A = \left[\frac{2X^{3/2}}{3} - \frac{X^3}{3} \right]^{1/4}$$

$$A = \left[\frac{2}{3} - \frac{1}{3} \right] - \left[\frac{2(1/4)^{3/2}}{3} - \frac{(1/4)^3}{3} \right]$$

$$= \frac{1}{3} - \left[2 - \left(\frac{1}{4} \right)^{3/2} \right] \frac{(1/4)^{3/2}}{3}$$

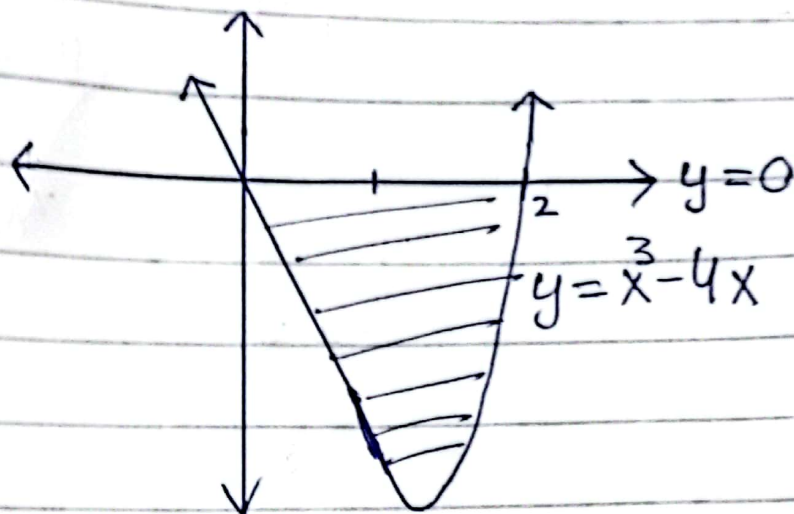
$$= \frac{1}{3} - \left[2 - \frac{1}{8} \right] \frac{1}{24}$$

$$= \frac{1}{3} - \left[\frac{15}{8} \right] \left(\frac{1}{24} \right)$$

$$= \frac{1}{3} - \left[\frac{5 \times 1}{8 \times 8} \right]$$

$$= \frac{1}{3} - \frac{5}{64} = \frac{49}{192}$$

8) $y = x^3 - 4x$, $y = 0$, $x = 0$, $x = 2$



Along x-axis:

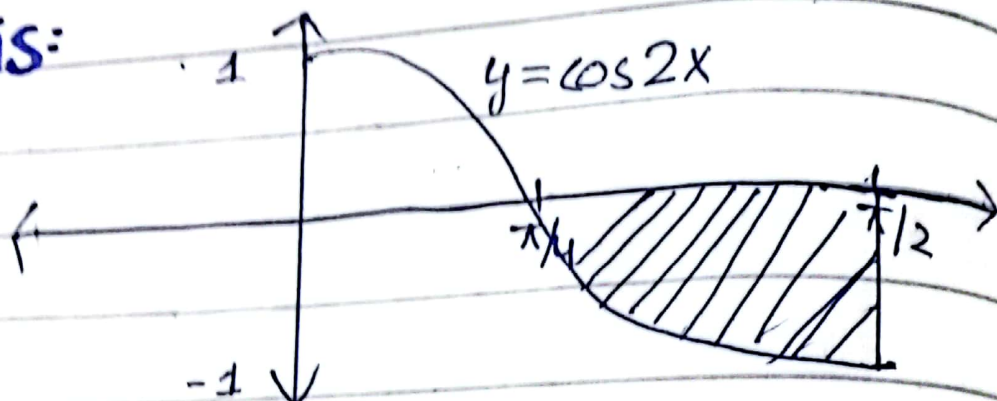
$$\begin{aligned}
 A &= \int_{x=0}^{x=2} [0 - (x^3 - 4x)] dx = \int_0^2 (4x - x^3) dx \\
 &= \left(\frac{4x^2}{2} - \frac{x^4}{4} \right) \Bigg|_0^2 \\
 &= \left(2x^2 - \frac{x^4}{4} \right) \Bigg|_0^2 \\
 &= 2(2)^2 - \frac{16}{4}
 \end{aligned}$$

$$A = 8 - 4$$

$$A = 4 \text{ sq. units}$$

9) $y = \cos 2x$, $y=0$, $x=\pi/4$, $x=\pi/2$

Along x-axis:



$$A = \int_{\pi/4}^{\pi/2} [0 - \cos 2x] dx$$

$$= - \int_{\pi/4}^{\pi/2} \cos 2x$$

$$= - \left[\frac{1}{2} \sin 2x \right]_{\pi/4}^{\pi/2}$$

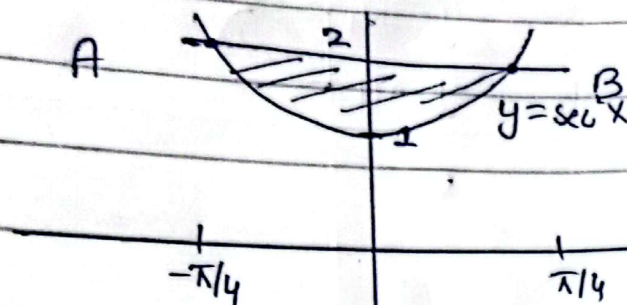
$$= - \frac{1}{2} \sin \pi + \frac{1}{2} \sin \frac{\pi}{2}$$

$$= 0 + \frac{1}{2}$$

$$A = 1/2$$

$$10) y = \sec^2 x, y = 2, x = -\pi/4$$

$$x = \pi/4$$



Along x-axis

For point A and B

$$2 = \sec^2 x$$

$$\sec x = \pm \sqrt{2}$$

$$x = \pm \pi/4$$

$$A = \int_{x=-\pi/4}^{x=\pi/4} (2 - \sec^2 x) dx$$

$$= [2x - \tan x]_{-\pi/4}^{\pi/4}$$

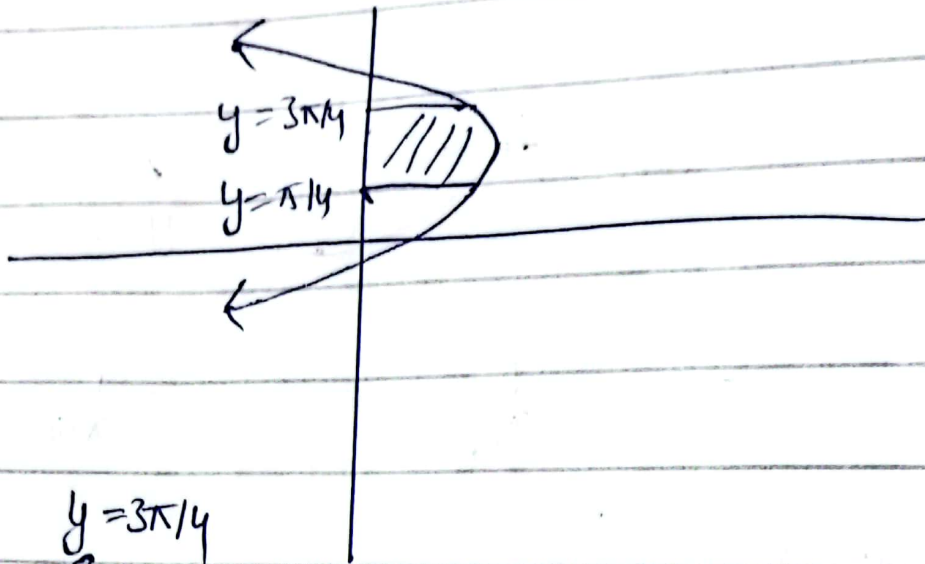
$$= [\pi/2 - \tan \pi/4] - [-\pi/2 - \tan(-\pi/4)]$$

$$= [\pi/2 - 1] - [-\pi/2 + 1]$$

$$= \pi/2 + \pi/2$$

$$= \pi$$

11) $x = \sin y, x=0, y=\pi/4, y=3\pi/4$



$$A = \int_{y=\pi/4}^{y=3\pi/4} \sin y \, dy$$

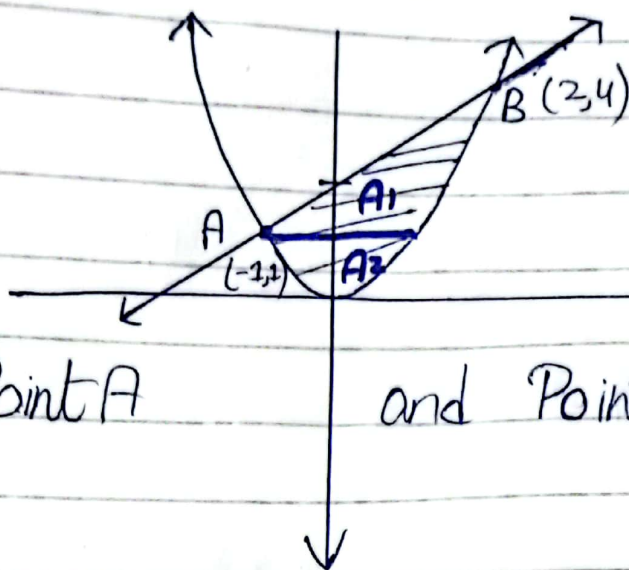
$$= -\cos y \Big|_{\pi/4}^{3\pi/4}$$

$$= -\cos 3\pi/4 + \cos \pi/4$$

$$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}}$$

$$= \sqrt{2}$$

$$12) x^2 = y, x = y - 2$$



For Point A and Point B, we get

$$x^2 = x + 2$$

$$x^2 - x - 2 = 0$$

$$x^2 - 2x + x - 2 = 0$$

$$x(x-2) + 1(x-2) = 0$$

$$(x-2)(x+1) = 0$$

$$x = 2$$

$$x = -1$$

$$B(2, 4)$$

$$A(-1, 1)$$

Along x-axis

$$A = \int_{x=-1}^{x=2} [x+2-x^2] dx$$

$$= \left(-\frac{x^3}{3} + \frac{x^2}{2} + 2x \right) \Big|_{-1}^2$$

$$= \left[-\frac{8}{3} + 2 + 4 \right] - \left[\frac{1}{3} + \frac{1}{2} - 2 \right]$$

$$= \frac{10}{3} + \frac{7}{6} = \frac{9}{2}$$

Along y-axis:

$$x = \sqrt{y}$$

$$x = y - 2$$

$$A_T = A_1 + A_2 \quad \text{--- (i)}$$

$$A_1 = \int_{y=1}^{y=4} (\sqrt{y} - y + 2) dy$$

$$= \left[\frac{2y^{3/2}}{3} - \frac{y^2}{2} + 2y \right]_1^4$$

$$= \frac{2(4)^{3/2}}{3} - \frac{16}{2} + 8 - \left[\frac{2(1)^{3/2}}{3} - \frac{1}{2} + 2 \right]$$

$$= \frac{16}{3} - 8 + 8 - \frac{2}{3} + \frac{1}{2} - 2$$

$$A_1 = 19/6$$

$$A_2 = \int_{y=0}^{y=1} 2\sqrt{y}$$

$$= \left[\frac{2y^{3/2}}{3/2} \right]_0^1$$

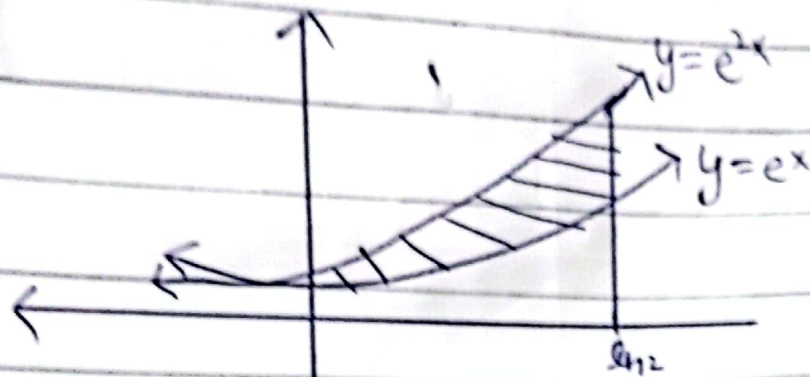
$$= \frac{4}{3} y^{3/2} \Big|_0^1$$

$$= \frac{4}{3} [1 - 0]$$

$$A_2 = 4/3$$

$$A_T = 19/6 + 4/3 = 9/2$$

13) $y = e^x$; $y = e^{2x}$; $x = 0$; $x = \ln 2$



Along X-axis

$$= \int_{x=0}^{x=\ln 2} (e^{2x} - e^x) dx$$

$$= \left[\frac{1}{2} e^{2x} - e^x \right]_0^{\ln 2}$$

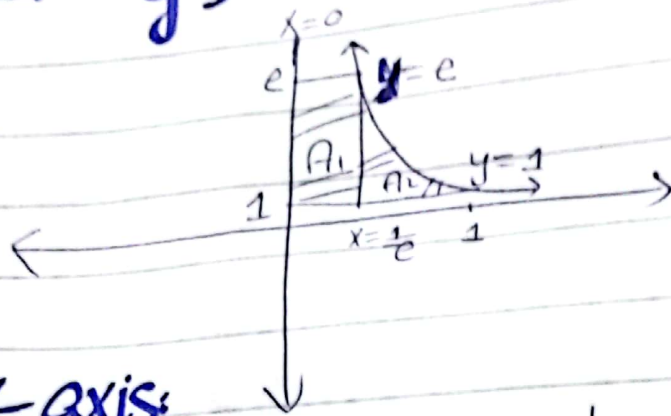
$$= \frac{1}{2} e^{2\ln 2} - e^{\ln 2} - \frac{1}{2} e^0 + e^0$$

$$= \frac{1}{2} e^{\ln 4} - e^{\ln 2} - \frac{1}{2} + 1$$

$$= \frac{4}{2} - 2 - \frac{1}{2} + 1$$

$$= 1/2$$

14) $x = \frac{1}{y}, x = 0, y = 1, y = e$



Along x-axis:

$$y = e$$

$$e = \frac{1}{x}$$

$$x = \frac{1}{e}$$

$$y = \frac{1}{x}$$

$$A_T = A_1 + A_2$$

$$A_1 = \int_{x=0}^{x=1/e} [e - 1] dx$$

$$= (e-1)x \Big|_0^{1/e}$$

$$A_1 = (e-1)\left(\frac{1}{e}\right)$$

$$A_2 = \int_{x=1/e}^{x=1} \left[\frac{1}{x} - 1 \right] dx$$

$$= (\ln|x| - x) \Big|_{1/e}^1$$

$$= \ln 1 - 1 - \ln \frac{1}{e} + \frac{1}{e}$$

$$= 0 - 1 - (-1)(1) + \frac{1}{e}$$

$$= -1 + 1 + \frac{1}{e} = \frac{1}{e}$$

$$H = \frac{1}{e} + \frac{1}{e}(e-1)$$

$$= \frac{1}{e}(1+e-1)$$

$$A_T = e/e = 1$$

Along y-axis

$$A = \int_{y=1}^{y=e} \frac{1}{y} dy$$

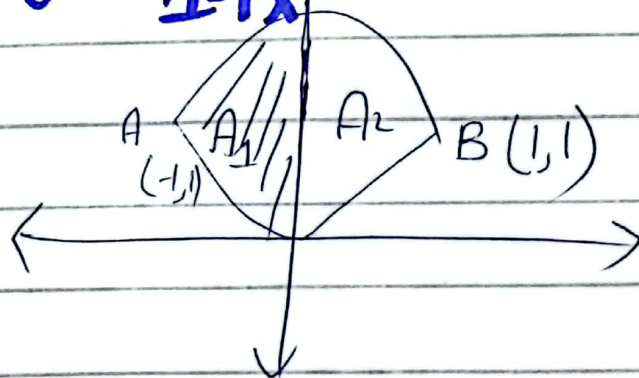
$$= \ln|y| \Big|_1^e$$

$$= \ln e - \ln 1$$

$$= 1 - 0$$

$$A = 1$$

15) $y = \frac{2}{1+x^2}$, $y = |x|$



$$\frac{2}{1+x^2} = x$$

$x = 1, -1$
are roots of
equation

$$2 = x(1+x^2)$$

$$= x + x^3 - 2$$

$$0 = x^3 + x - 2$$

$$A_1 = \int_{x=-1}^{x=0} \left[\frac{2}{1+x^2} - (-x) \right] dx$$

$$= \int_{x=-1}^{x=0} \left[\frac{2}{1+x^2} + x \right] dx$$

$$A_1 = \left[2 \tan^{-1} x + \frac{x^2}{2} \right]_{-1}^0$$

$$= 0 - \left[2 \tan^{-1}(-1) + \frac{1}{2} \right]$$

$$= -2 \tan^{-1}(-1) - \frac{1}{2}$$

$$= -2(-\pi/4) - 1/2$$

$$A_1 = \pi/2 - 1/2$$

$$A_2 = \int_{x=0}^{x=1} \left[\frac{2}{1+x^2} + x \right] dx$$

$$= \left[2 \tan^{-1} x + \frac{x^2}{2} \right]_0^1$$

$$= 2 \tan^{-1}(1) + \frac{1}{2} - 0$$

$$= 2(\pi/4) + \frac{1}{2}$$

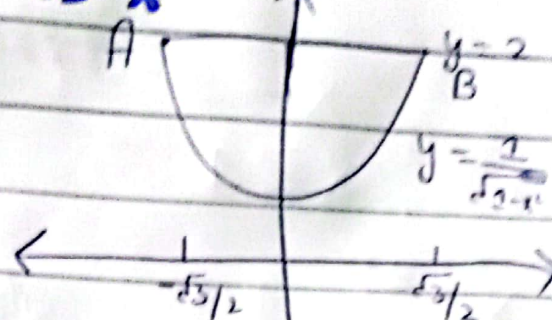
$$= \pi/2 + \frac{1}{2}$$

$$A_T = A_1 + A_2$$

$$= \frac{\pi}{2} + \frac{1}{2} + \frac{\pi}{2} + \frac{1}{2}$$

$$A_T = \pi + 1$$

$$16) \quad y = \frac{1}{\sqrt{1-x^2}}, \quad y=2$$



Along $\downarrow$ x-axis:

$$2 = \frac{1}{\sqrt{1-x^2}}$$

$$4 = \frac{1}{1-x^2}$$

$$4 - 4x^2 = 1$$

$$4x^2 = 3$$

$$x = \pm \sqrt{3}/2$$

$$A = \int_{-\sqrt{3}/2}^{\sqrt{3}/2} \left(2 - \frac{1}{\sqrt{1-x^2}} \right) dx$$

$$= \left(2x - \sin^{-1} x \right) \Big|_{-\sqrt{3}/2}^{\sqrt{3}/2}$$

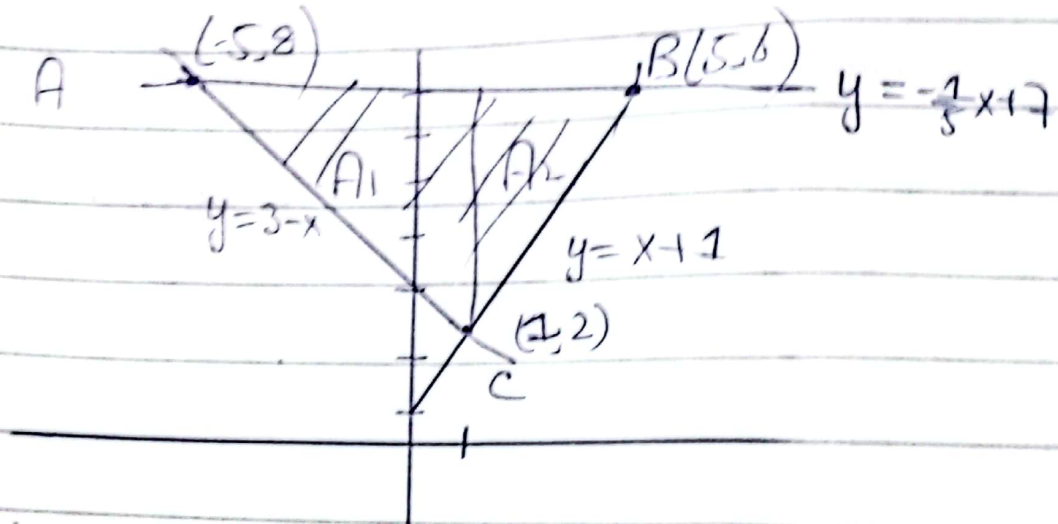
$$= \sqrt{3} - \sin^{-1}(\sqrt{3}/2) - \left[-\sqrt{3} - \sin^{-1}(-\sqrt{3}/2) \right]$$

$$= \sqrt{3} - \frac{\pi}{3} + \sqrt{3} + \frac{\pi}{3}$$

$$= 2\sqrt{3} - 2\pi/3$$

17) $y = 2 + |x-1|$, $y = -\frac{1}{5}x + 7$
Along x-axis

$$y = \begin{cases} 3-x & x < 1 \\ 1+x & x \geq 1 \end{cases}$$



Point A: $3 - x = -\frac{1}{5}x + 7$
 $3 - 7 = -\frac{1}{5}x + x$
 $-4 = \frac{4x}{5}$
 $-20 = 4x$
 $x = -5$

$A(-5, 8)$

Point B:

$$\begin{aligned} x + 1 &= -\frac{1}{5}x + 7 \\ x + x &= 5 \cdot 6 \\ 5x + 1 &= 30x \\ 5x^2 - 30x + 1 &= 0 \\ \frac{6x}{5} &= 6 \\ x &= 5 \\ B(5, 6) \end{aligned}$$

$$\begin{aligned} 3-x &= x+1 \\ 2x &= 2 \\ x &= 1 \end{aligned}$$

$$C(1, 2)$$

$$A_T = A_1 + A_2$$

$$\begin{aligned} A_1 &= \int_{x=-5}^{x=1} \left[\left(-\frac{x}{5} + 7 \right) - (3-x) \right] dx \\ &= \int_{-5}^1 \left(4 - \frac{x}{5} + x \right) dx \\ &= \int_{-5}^1 \left(4 + \frac{4x}{5} \right) dx \\ &= \left(4x + \frac{4x^2}{10} \right) \Big|_{-5}^1 \\ &= 4 + \frac{4}{10} - \left[4(-5) + \frac{4(-5)^2}{10} \right] \\ &= 4 + \frac{4}{10} - [-20 + 10] \\ &= 4 + \frac{4}{10} + 10 \\ &= 14 + \frac{2}{5} = \frac{72}{5} \end{aligned}$$

$$\begin{aligned} A_2 &= \int_{x=1}^{x=5} \left[\left(-\frac{x}{5} + 7 \right) - (x+1) \right] dx \\ &= \int_1^5 \left[6 - \frac{x}{5} - x \right] dx \\ &= \int_1^5 \left[6 - \frac{6x}{5} \right] dx \end{aligned}$$

$$= \int_1^5 \left[6 - \frac{6x}{5} \right] dx$$

$$= \left[6x - \frac{6x^2}{10} \right]_1^5$$

$$= \left[30 - 15 \right] - \left[6 - \frac{3}{5} \right]$$

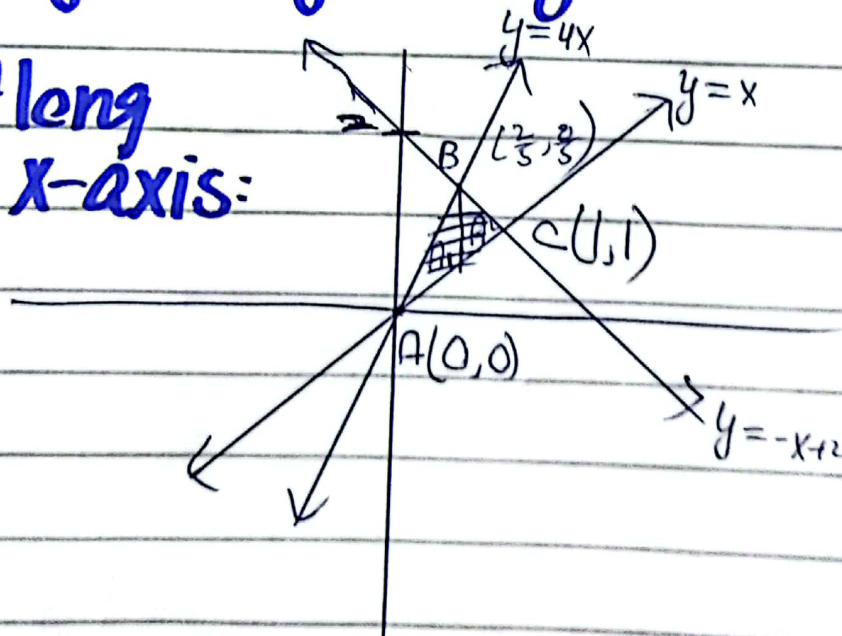
$$= 15 - \left[\frac{27}{5} \right]$$

$$= \frac{75 - 27}{5} = \frac{48}{5}$$

$$A_1 = \frac{72}{5} - \frac{48}{5} = \frac{24}{5} = 24$$

12) $y=x, y=4x, y=-x+2$

Along
x-axis:



Point **B**:

$$-x+2=x$$

$$2x=2$$

$$x=1 \quad C(1,1)$$

$$y = -x + 2$$

$$x = 2$$

$$x = \frac{2}{5}$$

$$B\left(\frac{2}{5}, \frac{2}{5}\right)$$

$$A_T = A_1 + A_2$$

$$A_1 = \int_0^{2/5} (4x - x) dx = \int_0^{2/5} 3x dx = \left[\frac{3x^2}{2} \right]_0^{2/5} = \frac{6}{25}$$

$$A_2 = \int_{2/5}^1 [-x + 2 - x] dx$$

$$= \int_{2/5}^1 (-2x + 2) dx$$

$$= \left[-x^2 + 2x \right]_{2/5}^1$$

$$= -x^2 + 2x \Big|_{2/5}^1$$

$$= -1 + 2 + \left(\frac{2}{5} \right)^2 - 2 \left(\frac{2}{5} \right)$$

$$= 1 + \frac{4}{25} - \frac{4}{5}$$

$$= \frac{25 + 4 - 20}{25} = \frac{9}{25}$$

$$A_T = \frac{6}{25} + \frac{9}{25} = \frac{15}{25} = \frac{3}{5}$$

Along y-axis

$$A_T = \int_{y=0}^{y=1} \left[y - \frac{1}{4}y \right] dy + \int_{y=1}^{y=2/5} \left[(2-y) - \frac{1}{4}y \right] dy$$

$$= \int_0^1 \left(y - \frac{y}{4} \right) dy + \int_1^{2/5} \left(2 - y - \frac{1}{4}y \right) dy$$

$$= \left(\frac{y^2}{2} - \frac{y^2}{8} \right) \Big|_0^1 + \left(2y - \frac{y^2}{2} - \frac{y^2}{8} \right) \Big|_1^{2/5}$$

$$= \frac{1}{2} - \frac{1}{8} + \left[\frac{16}{5} - \frac{64}{50} - \frac{64}{200} \right]$$

$$- \left[2 - \frac{1}{2} - \frac{1}{8} \right]$$

$$= \frac{3}{8} + \frac{8}{5} - \frac{11}{8}$$

$$= \frac{8}{5} - 1 = \frac{3}{5}$$